

Detecting Gender Bias in Occupational Word Embeddings

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Abstract

This study investigates societal gender biases encoded in pre-trained word embeddings, specifically focusing on occupational terms. Using the Google News Word2Vec and GloVe 6B (300d) models, we employ multiple bias detection methods, including gender direction projection, cosine similarity, analogy tests, and the Word Embedding Association Test (WEAT). Our analysis reveals significant gender associations in embeddings that align with societal stereotypes, such as strong female associations for roles like "nurse" and "midwife," and male associations for "engineer" and "chief." These biases correlate moderately with real-world gender distributions from the U.S. Bureau of Labor Statistics (BLS), highlighting the reflection of societal norms in AI models. We also explore a simple debiasing technique to mitigate these biases. The findings underscore the importance of addressing embedded biases in AI systems to prevent the perpetuation of stereotypes in applications like hiring or career recommendation.

1. Introduction

Artificial intelligence (AI) models, particularly those based on natural language processing (NLP), are increasingly used in decision-making processes, including hiring, content recommendation, and career guidance. However, these models can inadvertently encode societal biases present in their training data, leading to discriminatory outcomes. This project focuses on detecting gender biases in occupational contexts within pre-trained word embeddings—vector representations of words that capture semantic relationships. By analyzing embeddings from the Google News Word2Vec and GloVe models, we aim to quantify and understand the extent to which these models reflect societal stereotypes about gendered jobs.

Understanding and mitigating such biases is crucial for developing fair and equitable AI systems. This report presents a comprehensive analysis of gender bias in occupational word embeddings, correlates these biases with real-world data, and explores a preliminary debiasing approach.

2. Methodology

2.1 Word Embedding Models

We utilized two widely used pre-trained word embedding models:

- **Google News Word2Vec:** A 300-dimensional model trained on Google News articles.

- **GloVe 6B (300d)**: A 300-dimensional model trained on a combination of Wikipedia and Gigaword corpora.

These models were chosen for their broad coverage and established use in bias detection research.

2.2 Bias Detection Techniques

To quantify gender bias in occupational terms, we employed the following methods:

1. Gender Direction Projection:

- A gender direction vector was defined using pairs of gendered words (e.g., "man - woman," "he - she"). For GloVe, multiple pairs were averaged to create a robust gender axis.
- Occupational vectors were projected onto this gender direction to compute bias scores, where positive scores indicate male association and negative scores indicate female association.

2. Cosine Similarity:

- We calculated the cosine similarity between occupational vectors and gender terms (e.g., "man," "woman") to measure direct associations.

3. Analogy Tests:

- We used analogy patterns like "man is to occupation as woman is to ?" to reveal stereotypical completions encoded in the embeddings.

4. Word Embedding Association Test (WEAT):

- WEAT was applied to measure the differential association between stereotypically male and female occupations and gender attributes (e.g., "male," "female").

2.3 Data

- **Occupational Terms**: A list of 382 occupations was sourced from the U.S. Bureau of Labor Statistics (BLS), providing real-world gender distribution data (percentage of women employed).
- **Handling Multi-Word Occupations**: For occupations not directly present in the vocabulary (e.g., "social worker"), we used variants or individual words (e.g., "social," "worker") where applicable.

3. Results

3.1 Gender Association Scores

Using the gender direction projection method, we computed bias scores for occupational terms. Key findings include:

- **Strong Female Associations:**
 - midwife (-1.2599), nurse (-1.0206), flight_attendant (-1.1711)
- **Strong Male Associations:**
 - general (1.2637), chief (1.0361), packers (1.0342)

These scores reflect societal stereotypes, with caregiving roles showing female bias and leadership or manual labor roles showing male bias.

3.2 Cosine Similarity Analysis

Cosine similarity results reinforced the projection findings:

- nurse: Similarity to "woman" (0.4414) > "man" (0.2547)
- engineer: Similarity to "man" (0.1513) > "woman" (0.0944)

3.3 Correlation with Real-World Data

We correlated the bias scores with BLS data on the percentage of men in each occupation:

- **Pearson Correlation:** 0.3546 ($p < 0.0001$), indicating a moderate but statistically significant relationship between embedding biases and real-world gender distributions.

3.4 WEAT Analysis

The WEAT analysis yielded:

- **Effect Size:** 0.2167
- **P-value:** 0.6490 (non-significant), suggesting weak differential association, possibly due to noisy term sets.

3.5 Debiasing Evaluation

A soft debiasing method (reducing the gender component by 50%) was applied to occupational vectors:

- **Bias Reduction:** Consistent 50% reduction in bias scores across occupations.
- **Limitation:** No evaluation of semantic utility post-debiasing.

4. Discussion

4.1 Implications of Findings

The results demonstrate that word embeddings encode societal gender stereotypes, with occupations like "nurse" and "engineer" reflecting strong female and male associations, respectively. The moderate correlation with BLS data suggests that while embeddings capture real-world trends, they also amplify stereotypes, as seen in cases like "doctor" showing female bias despite being 41.4% female in reality.

These biases can have significant implications for AI applications, such as:

- **Hiring Systems:** Biased embeddings could favor certain genders for specific roles.
- **Career Recommendations:** Embeddings might suggest stereotypical career paths based on gender.

4.2 Limitations

- **Multi-Word Occupations:** Handling of multi-word terms (e.g., "social worker") was suboptimal, often relying on individual words like "social," which may not fully capture the occupation's semantics.
- **Gender Direction:** The use of a single gender pair in Word2Vec may oversimplify the gender axis.
- **WEAT Results:** The inclusion of ambiguous terms (e.g., "and," "other") weakened the WEAT analysis.
- **Debiasing:** The debiasing method was simplistic and did not assess the preservation of semantic meaning.

5. Conclusion

This project successfully quantified gender biases in occupational word embeddings, revealing that AI models reflect and potentially amplify societal stereotypes. The correlation between embedding biases and real-world data underscores the need for careful consideration of bias in AI applications. While our debiasing approach reduced bias, future work should explore more sophisticated methods and evaluate their impact on downstream tasks.